From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2013

FULL TEST – I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
   Section – A (05 – 09) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 mark** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.
3. **Section – C (01 – 08)** contains 8 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

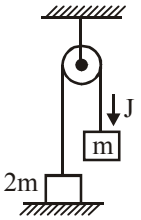
Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics PART – I

SECTION – A
Single Correct Choice Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In the given figure an impulse J is given to the block of mass m in the downward direction. As a result of the impulse



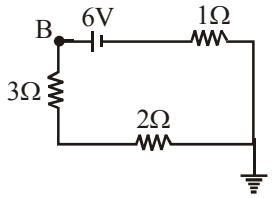
(A) both the blocks start moving with $\frac{J}{3m}$ in opposite directions

(B) both the blocks start moving with $\frac{J}{m}$ in opposite direction

(C) the centre of mass of the system is moving downwards

(D) the centre of mass of the system is not moving

2. In the given electrical circuit electric Potential of point B is



(A) 6 VOLT (B) 5 VOLT

(C) 4 VOLT (D) 3 VOLT

3. Choose the incorrect statement about the coefficient of restitution

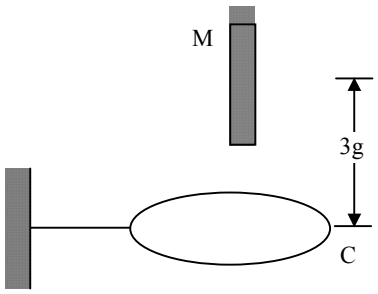
(A) it is defined as the ratio of relative velocity of separation to the relative velocity of approach

(B) it is defined only for two colliding bodies

(C) it is defined for colliding parts of the smooth body along the normal to the striking surface

(D) while calculating the velocity of approach and separation we take the velocities of centre of mass of the bodies

4. A bar magnet M is allowed to fall towards a fixed conducting ring C. If g is the acceleration due to gravity, v is the velocity of the magnet at t = 2s and s is the distance traveled by it in the same time then



(A) v > 2g (B) v < 4g

(C) s > 2g (D) s < 2g

Rough work

Multiple Correct Answer(s) Type

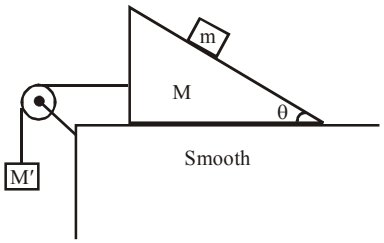
This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

5. Which of the following are true for a satellite in an orbit.
   (A) it is a freely falling body
   (B) its velocity is constant
   (C) it suffers no acceleration
   (D) it does not require energy for its motion in the orbit

6. The molar heat capacity for an ideal gas
   (A) is zero for an adiabatic process
   (B) is infinite for an isothermal process
   (C) depends only on the nature of the gas for a process in which either volume or pressure is constant
   (D) is equal to the product of the molecular weight and specific heat capacity for any process

7. A particle is projected at an angle $\theta = 30^\circ$ with the horizontal, with a velocity of 10 m/s then
   (A) after 2 s the velocity of particle makes an angle of 60º with initial velocity vector
   (B) after 1 s the velocity of particle makes an angle of 60º with initial velocity vector
   (C) the magnitude of velocity of particle after 1 s is 10 m/s
   (D) the magnitude of velocity of particle after 1 s is 5 m/s

8. The figure shows a block of mass *m* placed on a smooth wedge of mass *M*. Calculate the value of M′ and tension in the string, so that the block of mass *m* will move vertically downward with acceleration 10 m/s^2. (Take g = 10 m/s^2)



   (A) the value of M′ is $\frac{M\cot\theta}{1-\cot\theta}$

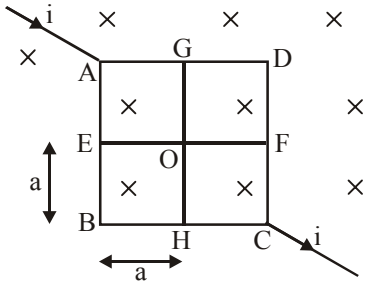
   (B) the value of M′ $\frac{M\tan\theta}{1-\tan\theta}$

   (C) the value of tension in the string is $\frac{Mg}{\tan\theta}$

   (D) the value of tension is $\frac{Mg}{\cot\theta}$

Rough work

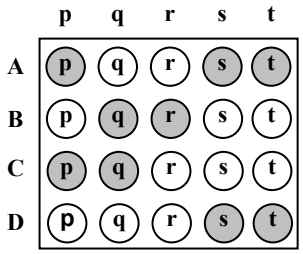
9. In the figure, there is a uniform conducting structure in which each small square has side a. The structure is kept in uniform magnetic field B.



(A) The magnetic force on the structure is $2\sqrt{2}$ iBa .
(B) The potential of point B = potential of point D.
(C) Potential of point O = potential of point B.
(D) The magnetic force on the structure is $\sqrt{2}$ iBa .

SECTION - B
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. A real object is being seen by optical component listing in column A and nature of image of this object is listing in column B :

Column A	**Column B** **Nature of image**
(A) Convex lens.	(p) Real.
(B) Convex mirror.	(q) Virtual.
(C) Concave lens.	(r) Erect.
(D) Concave mirror.	(s) Inverted.

2. Column A lists type of decay and the column B lists reason of decay (reason for instability of nucleus)

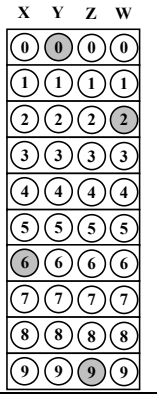
Column A	**Column B**
(A) γ–decay.	(p) Nucleus is too large.
(B) α–decay.	(q) Nucleus has too many neutrons relative to number of protons.
(C) β minus decay.	(r) Nucleus has excess energy.
(D) β plus decay.	(s) Nucleus has too many protons relative to number of neutrons.

Rough work

SECTION – C

Integer Answer Type

This section contains 8 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.

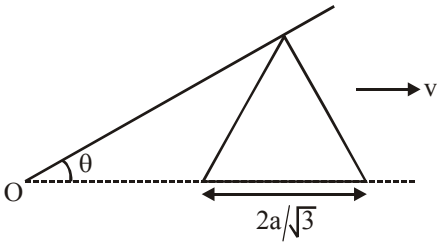


1. A body of mass m = 4 kg starts moving with velocity v_0 in a straight line is such a way that on the body work is being done at the rate which is proportional to the square of velocity as given by $P = \beta v^2$ where $\beta = \frac{0.693}{2}$. Find the time elapsed in seconds before velocity of body is doubled.

2. A sound source of frequency f_0 = 130 Hz is dropped from a height slightly greater than 250 m above the ground. At the same time a detector is thrown upwards with velocity u = 50 ms^{-1} along the same line. If the speed of sound is v = 300 ms^{-1}, if the frequency (in Hz) detected by the detector after t = 5 s is 39 × n, find the value of n (Take g = 10 ms^{-2})

3. There is an isolated planet having mass 2M and radius 2R, where M and R are the mass and radius of the earth. A simple pendulum having mass m and length 2R is made to small oscillations on the planet. The time period of SHM of pendulum is (0.7 K + 0.5) seconds. Find the value of K. (take π = 3.00, g = 10 m/s^2, $\sqrt{2} = 1.41$)

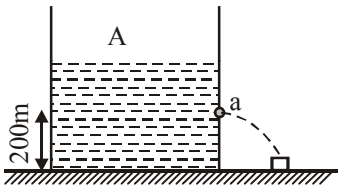
4. In the arrangement shown the rod is freely pivoted at point O and is in contact with the equilateral triangular block which can moves on the horizontal frictionless ground. As the block is given a speed v forward, the rod rotates about point O. Find the angular velocity of rod in rad/s at the instant when θ = 30º. [Take v = 20 m/s, a = 1 m]



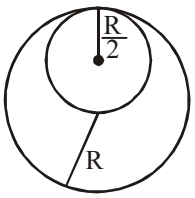
5. Radiation from hydrogen gas excited to first excited state is used for illuminating certain photoelectric plate. When the radiation from some unknown hydrogen like gas excited to the same level is used to expose the same plate, it is found that the de–Broglie wavelength of the fastest photoelectron has decreased 2.3 times. It is given that the energy corresponding to the longest wavelength of the Lyman series of the unknown gas is 3 times the ionization energy of hydrogen gas (13.6 eV). Find the work function of photoelectric plate in eV. (Take $(2.3)^2$ = 5.25.

Rough work

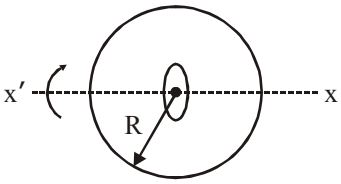
6. Water is filled in a uniform container of area of cross section A. A hole of cross section area a (<< A) is made in the container at a height of 20 m above the base. Water streams out and hits a small block placed at some distance from container. With what speed (in ms^{-1}) the block should be moved such that water streams always hits the block. (Given $\frac{a}{A}=\frac{1}{20}$). (Take g = 10 ms^{-2})



7. A spherical hole of radius $\frac{R}{2}$ is drilled from a planet of mass M as shown in the figure. If the gravitational acceleration at a point on the surface of the planet just above the hole is N × 1.1 m/s^2, find the value of N.
[Given : G = 6.6 × 10^{-11} $N.m^2/kg^2$, R = 6 × 10^6 m, M = 6 × 10^{24} kg]



8. A non–conducting ring of radius R having uniformly distributed charge Q starts rotating about x–x′ axis passing through diameter with an angular acceleration α as shown in the figure. Another small conducting ring having radius a (a << R) is kept fixed at the centre of bigger ring is such a way that axis xx′ is passing through its centre and perpendicular to its plane. If the resistance of small ring is r = 1Ω, find the induced current in it in ampere.
(Given $q=\frac{16\times10^2}{\mu_0}$C, R = 1 m, a = 0.1 m, α = 8 rad/s^2)



Rough work

Chemistry

PART – II

SECTION – A

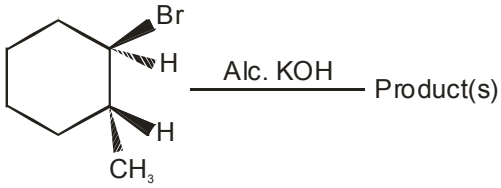
Straight Objective Type

This section contains 4 multiple choice questions numbered 1 to 4. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. A 75 gm $4\%\left(\frac{w}{w}\right)$ aqueous NaOH solution is distilled for some time. The water collected has a mass of 25 gms. To this distilled solution X gms of KOH(s) is added, so that molality of OH^- ion in the resulting solution becomes 5. Find X?

   (A) 7 (B) 9

   (C) 7.4 (D) 8.2

2. 

   The above reaction follows bimolecular paths. Find the major product of the following :

   (A) Saytzeff alkene (B) Hoffmann alkene

   (C) alcohol (secondary) (D) alcohol (tertiary)

3. A 1.025 gm sample containing a weak acid HX (mol. wt = 82) is dissolved in 60 ml H_2O and titrated with 0.25 M NaOH solution. When half of the acid is neutralized the pH was found to be 5.0 and the pH at equivalence point is 9.0. Calculate the percentage of HX in the original sample. (Assume law of conservation of volume is obeyed)

   (A) 40% (B) 80%

   (C) 65% (D) 78%

4. The half life of a radioactive substance is 20 minutes. The approximate time interval ($t_2 - t_1$) between the time t_2 when 2/3 of it has decayed and t_1 when 1/3 of it has decayed is :

   (A) 14 min (B) 20 min

   (C) 28 min (D) 17 min

Rough Work

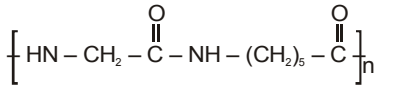
Multiple Correct Choice Type

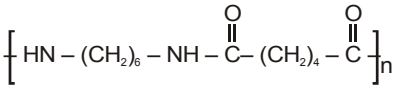
This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

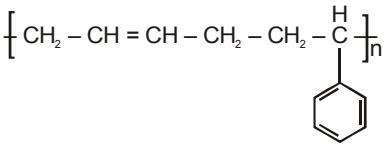
5. Find out which of the following statements are true :
   (A) Bothe pH and POH decreases with rise in temperature for pure H_2O.
   (B) pH decreases while POH increases with rise in temperature for pure H_2O.
   (C) The pH scale shortens with increase in temperature.
   (D) K_w increases with rise in temperature as dissociation of water is exothermic.

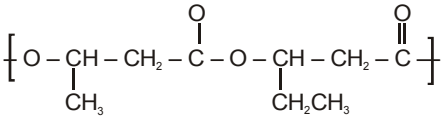
6. Which of the following statement(s) is(are) wrong ?
   (A) Hg_2Cl_2 in soluble in cold H_2O.
   (B) Hg_2Cl_2 dissolves in hot H_2O.
   (C) Hg_2Cl_2 gives a white ppt with NH_4OH.
   (D) Hg_2Cl_2 is used as a purgative.

7. Among the following, the heteropolymer is/are :

   (A) 

   (B) 

   (C) 

   (D) 

8. The observed rate of a chemical reaction is substantially lower than the collision frequency. One or more of the following statements is/are true to account for this fact.
   (A) the reactants do not have the required energy.
   (B) the partners do not collide in the proper orientation.
   (C) collision complex exists for a very short time.
   (D) collision frequency overestimates the number of effective collisions.

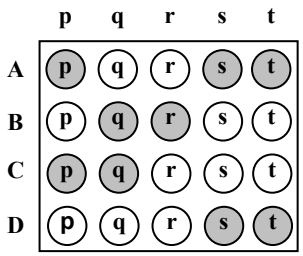
9. 5 ml of 0.5 M K_2SO_4 is added to 10 ml of 1 M $BaCl_2$ solution. What will happen?
   (A) Freezing point will increase
   (B) Boiling point will increase
   (C) Freezing point will decrease
   (D) Boiling point will decrease

Rough Work

SECTION-B
(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following and write the correct pairs.

Column – I (Polymer)			Column – II
(A)	Terylene	(p)	Poly amide
(B)	Nylon-6,6	(q)	Addition polymer
(C)	Starch	(r)	Condensation polymer
(D)	Teflon	(s)	Polyester
		(t)	Natural polymer

2. Match the following and write the correct pairs.

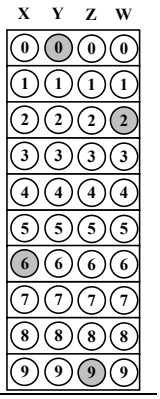
Column – I			Column – II
(A)	H_3PO_3	(p)	Mono basic
(B)	H_3PO_4	(q)	Dibasic
(C)	$H_4P_2O_7$	(r)	Disproportionation
(D)	H_3PO_2	(s)	Oxidation state of P is +5
		(t)	sp^3 hybridization

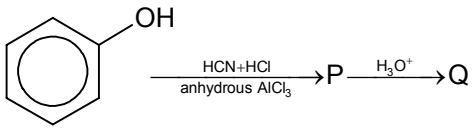
Rough Work

SECTION – C

Integer Answer Type

This section contains 8 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. 

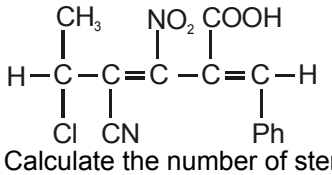
Find the degree of unsaturation in P.

2. In the reaction $[\underset{(CiS)}{CoCl_2(NH_3)_4}]^+ + Cl^- \rightarrow [CoCl_3(NH_3)_3] + NH_3$. How many isomers of the product (co-complex) is obtained.

3. One mole of Borax is dissolved in H_2O and the solution volume is made up to 2L. Calculate the minimum number of moles of NaOH required to neutralize the solution.

4. A compound AB has a rock salt structure with A : B = 1 : 1. The formula weight of AB is 6.023 Y amu. And the closest distance between A and B is $Y^{1/3}$ nm. The observed density of the lattice in kg/m³ is :

Rough Work

5. 
Calculate the number of stereoisomer possible for the above compound.

6. A vessel fitted with a weightless, frictionless piston of 0.025 m², contains excess conc. Of HCl. The piston moved 1 m outward when 0.075 kg iron fillings were added at 300 K. The solution left behind was found to contain Fe(II). The approximate purity of the iron sample is x × 10. Find x ?

7. Number of B – O bonds in diborate ion $\left(B_2O_5\right)^{2-}$ is

8. Benzaldehyde, on reaction with aniline forms Schiff's base. The number of π bonds in Schiff's base is

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 4 multiple choice questions numbered 1 to 4. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Consider a point $P(at^2, 2at)$ on the parabola $y^2 = 4ax$, A focal chord PS (S focus) is drawn to meet parabola again at Q. From Q, a normal is drawn to meet parabola again at R. From R, a tangent is drawn to the parabola to meet focal chord PSQ (extended) at T. The area of ΔQRT is

(A) $8a^2\left[t^2+\frac{1}{t^2}\right]^3$ (B) $a^2\left[t+\frac{1}{t}\right]^4$

(C) $\frac{8a^2}{3}\left[t+\frac{1}{t}\right]^3$ (D) $4a^2\left[t+\frac{1}{t}\right]^4$

2. Equation of plane which passes through the point of intersection of lines $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x-3}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ and at greatest distance from the point (0, 0, 0) is

(A) $4x + 3y + 5z = 25$ (B) $4x + 3y + 5z = 50$

(C) $3x + 4y + 5z = 49$ (D) $x + 7y – 5z = 2$

3. If $a > b > c$ and the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$ and $cx + ay + bz = 0$ has a non–trivial solution, then the quadratic equation $ax^2 + bx + c = 0$ has

(A) at least one positive root (B) positive roots

(C) roots of opposite sight (D) image roots

4. Solution of the differential equation $\left(x\frac{dy}{dx}+y\right)=e^{xy-\ln x^2}\left(x\frac{dy}{dx}-y\right)$

(A) $\frac{y}{x}-e^{-xy}=c$ (B) $\frac{x}{y}+e^{-xy}=c$

(C) $\frac{y}{x}+e^{-xy}=c$ (D) $-\frac{x}{y}+e^{-xy}=c$

Rough work

Multiple Correct Choice Type

This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

5. All the three roots of $az^3 + bz^2 + cz + d = 0$ have negative real parts (a, b, c, $\in$ R, $a \neq 0$) then
(A) ab > 0 (B) bc > 0
(C) ad > 0 (D) a > 0

6. If α and β^2 are the roots of $8x^2 - 10x + 3 = 0$ then the equation whose roots are $(\alpha + i\beta)^{100}$ and $(\alpha - i\beta)^{100}$ can be
(A) $x^2 + x + 1 = 0$ (B) $x^2 - x + 1 = 0$
(C) $x^3 - 1 = 0$ (D) $2x^2 + x + 1 = 0$

7. If function $f(x) = \begin{cases} |x-2| & ; \ x \geq 1 \\ \frac{x^2}{5} - \frac{4x}{5} + \frac{8}{5} & ; \ x < 1 \end{cases}$
(A) continuous at x = 1 (B) differentiable at x = 1
(C) not differentiable at x = 1 (D) differentiable at x = 2

8. Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If three vectors $\vec{p}, \vec{q}, \vec{r}$ parallel to $\vec{a}, \vec{b}, \vec{c}$ respectively and having integral but different magnitudes, then among the following options $|\vec{p} + \vec{q} + \vec{r}|$ can take a value equal to
(A) 1 (B) 0
(C) $\sqrt{3}$ (D) 2

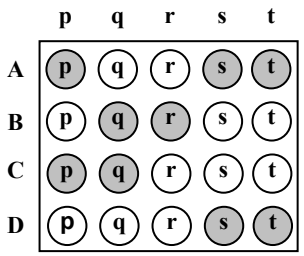
9. Let 'z' be a complex number and 'a' be a real parameter such that $z^2 + az + a^2 = 0$, then
(A) locus of z is a pair of straight lines (B) locus of z is a circle
(C) $\arg(z) = \pm\frac{2\pi}{3}$ (D) |z| =|a|

Rough work

SECTION – B

(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Given pair of lines $2x^2 + 5xy + 2y^2 + 4x + 5y + a = 0$ and line $L \equiv bx + y + 5 = 0$

	Column – I		Column – II
(A)	If there exist 4 circles which touch pair of lines and the line L simultaneously then the value of b can be	(p)	$\frac{1}{2}$
(B)	If there exist 2 circles which touch pair of lines and the line L simultaneously then the value of b can be	(q)	2
(C)	If there exist no circle which touches pair of lines and the line L simultaneously then the value of b can be	(r)	5
(D)	If there exist infinite circles which touch pair of lines and the line L simultaneously then the value of b can not be	(s)	4
		(t)	1

2. Match the following column–I with column–II.

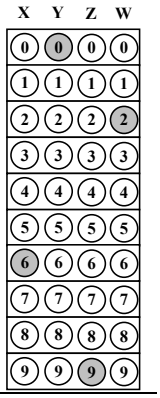
	Column – I		Column – II
(A)	Number of integral values of K such that all solutions (x, y) to the system of equation $x + 4y = 2K^2$ and $x + y = K$ are such that x, y > 0 is	(p)	0
(B)	Number of integers in the domain of function $g(x) = \ln(\cos^{-1} x)$ is	(q)	1
(C)	If x > 0; then the value of expression $\frac{(1+\sin^{-1}x)^{2010}(1+\cos^{-1}x)^{2011}(1+\tan^{-1}x)^{2012}}{(1+\text{cosec}^{-1}x)^{2010}(1+\sec^{-1}x)^{2011}(1+\cot^{-1}x)^{2012}}$ is	(r)	2
(D)	If the function $f : R - \{-b\} \to R - \{1\}$ defined by $f(x) = \frac{x+a}{x+b}$, $(a \neq b)$ is self inverse, then a + b can be	(s)	3
		(t)	4

Rough work

SECTION – C

Integer Answer Type

This section contains 8 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. If largest constant such that $\frac{Kabc}{a+b+c} \le (a+b)^2 + (a+b+4c)^2 \ \forall$ a, b, c > 0 is k then $\frac{K}{25}$ is equal to _____.

2. If y = f(x) satisfies f(x + 1) + f(z – 1) = f(x + z) $\forall$ x, z $\in$ R and f(0) = 0 and f′(0) = 4 then f(2) is equal to _____.

3. If $-\frac{\pi}{2} < \alpha_1 < \alpha_2 < \alpha_3 < \frac{\pi}{2}$, then number of values of $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $(\tan\theta - \tan\alpha_1)(\tan\theta - \tan\alpha_2)(\tan\theta - \tan\alpha_3) - \left(\sum(\tan\theta - \tan\alpha_i)\right)$ is _____.

4. A quadratic equation with integral coefficients has two prime numbers as its roots. If the sum of the coefficients of the equation is prime, then the sum of the roots is _____.

5. Let $I_1 = \int_0^1 \frac{e^x}{1+x}dx$ and $I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2-x^3)}dx$. The $\frac{I_1}{I_2 e}$ is equal to _____.

6. In a right angled triangle ABC with C as a right angle, a perpendicular CD is drawn to AB. The radii of the circles inscribed into the triangles ACD and BCD are equal to 3 and 4 respectively. Then the radius of the circle inscribed into the ΔABC is _____.

7. If $f(x) + f\left(1 - \frac{1}{x}\right) = 1 + x \ \forall\ x \in R - \{0, 1\}$. The value of 4f(2) is equal to _____.

8. Given A = $\cos^2\theta(\cos^2\theta + 1) + 2\sin^2\theta$. Then for all θ, [A] can assume values whose sum is _____ (Where [.] denotes the greatest integer function).

Rough work